Blockchain Technology (CSE4444)
Assignment 2 Submission Document

Student Name:	Aakarsh Jha
Registration Number:	230905248
Assignment Title:	Blockchain Technology (CSE4444) - Assignment 2
Title of Developed DApp:	CertiTrust: Decentralized Academic Credential Verification System
GitHub Repository Link:	https://github.com/Aakarsh-Jha/Blockchain-Assignment-2
Contract Address:	0xCfEB869F69431e42cdB54A4F4f105C19C080A601
Authority Wallet:	0x90F8bf6A479f320ead074411a4B0e7944Ea8c9C1

1. DApp Overview & Problem Statement

CertiTrust is an original decentralized application (DApp) designed to eliminate academic credential forgery. Degrees and certificates are cryptographically issued by an authorized university admin on the Ethereum blockchain. Each record stores the student name, degree program, reference hash, and an immutable block timestamp. Employers, verification agencies, and students can instantly query the smart contract to verify authenticity.

2. Input Validation & Test Cases Summary

Test Case	Caller	Inputs	Observed Outcome
TC-01 (Valid)	Owner	Valid Name & Course	Mined in Block #4; Certificate #1 issued successfully.
TC-02 (Invalid)	Owner	Empty student name ""	Reverted: 'Student name cannot be empty.'
TC-03 (Unauthorized)	Non-Owner	issueCertificate() call	Reverted: 'Unauthorized! Only authority can perform this action.'

3. Screenshots & Execution Evidence

Figure 1: Smart Contract Compilation (Remix IDE)

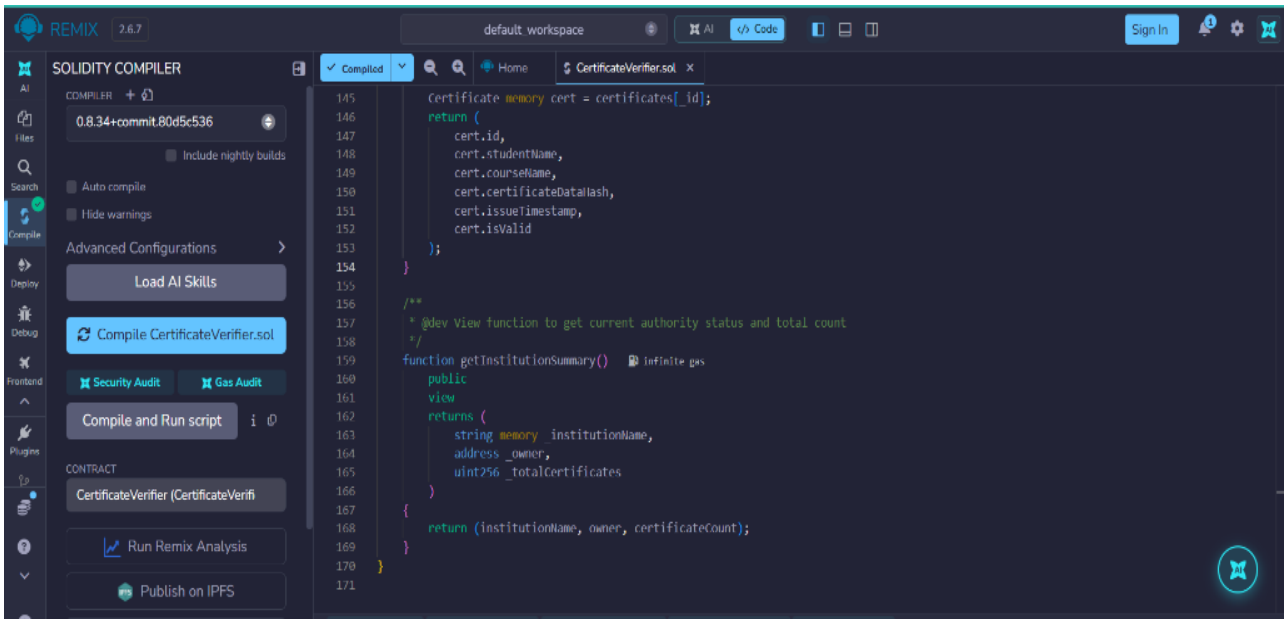


Figure 2: Smart Contract Deployment (Ganache Network)

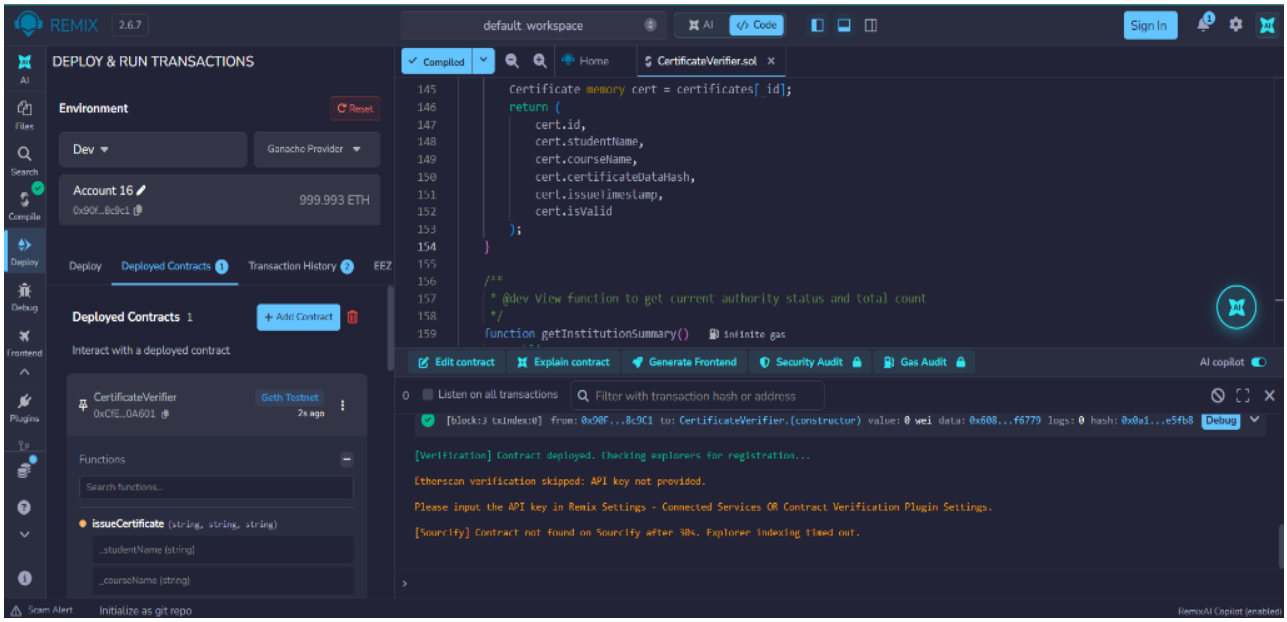


Figure 3: MetaMask Wallet Connection Handshake

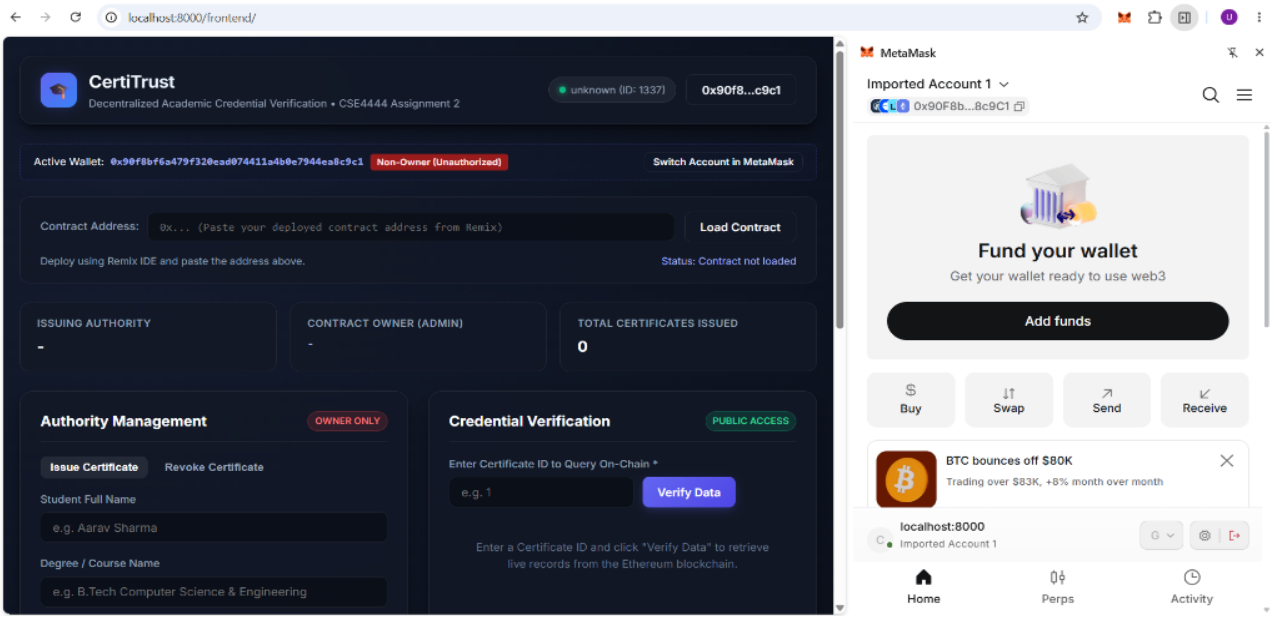


Figure 4: Successful Transaction Execution & Mining (Block #4)

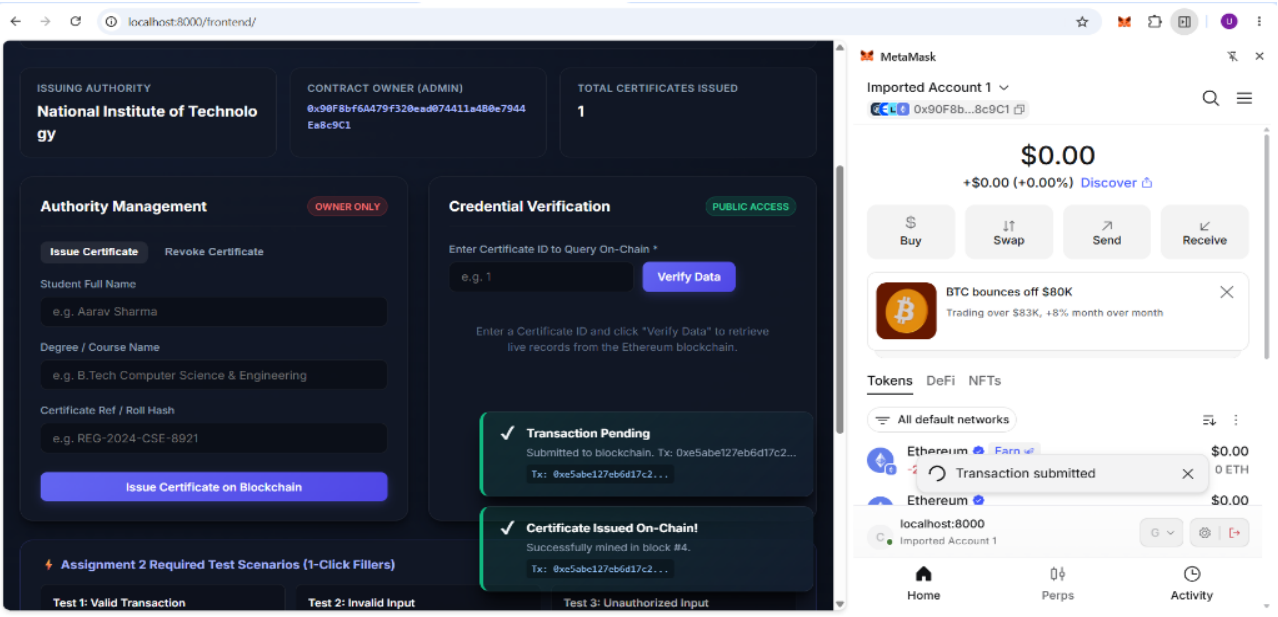


Figure 5: Failed Transaction Revert (Validation / Unauthorized)

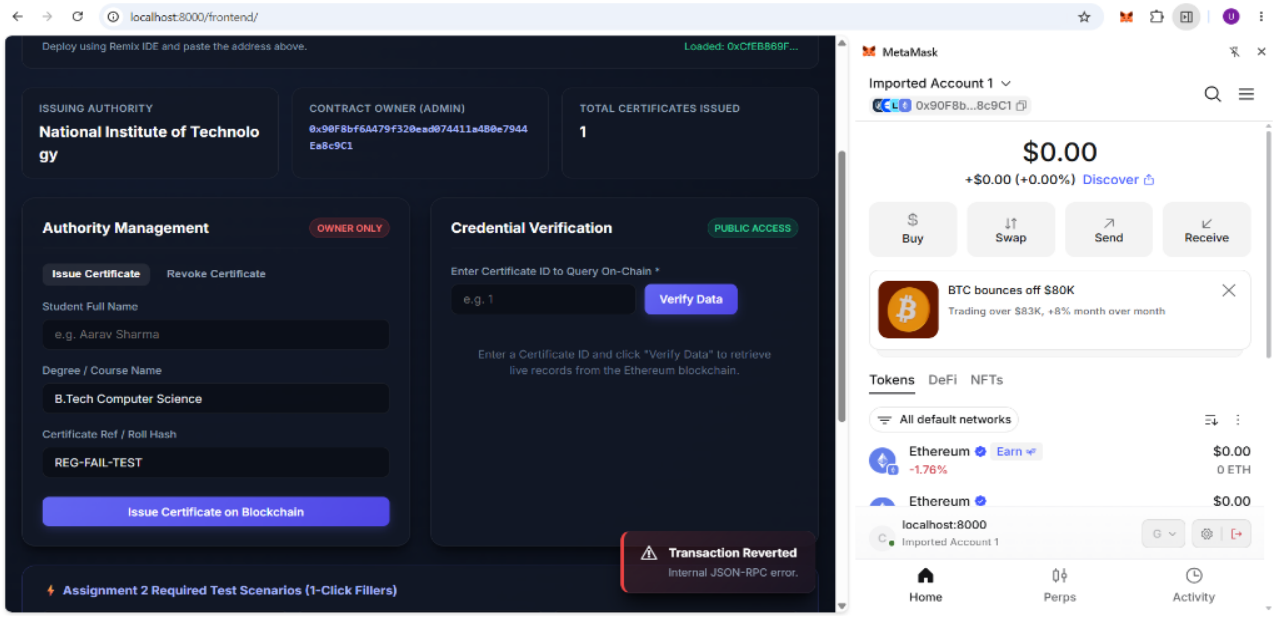


Figure 6: Blockchain Credential Retrieval & Verification in UI

